

Dynamic Allocation

Monday, May 18, 2026 10:06 AM

Usually in C, you use malloc to allocate memory on the heap and free to release it. In python this is handled via Garbage collection. When you create an object or a list, python allocates memory automatically. When no variables refer to that object anymore, Python deletes it automatically.

In C you usually use realloc to grow an array. Python's built-in list is a dynamic array, so you just use append.

```
# Slides 53-68: Reversing an array dynamically
```

```
def invert_order():
    n = int(input("N? "))
    v = [] # Dynamically sized array
    for i in range(n):
        val = float(input(f"El. {i}: "))
        v.append(val) # Equivalent to reallocating and inserting

    # Print in reverse
    for val in reversed(v):
        print(val)
```

```
# Slides 69-90: 2D Matrices
```

```
def create_matrix(rows, cols):
    # Equivalent to allocating an array of pointers, then an array for each row
    matrix = [[0.0 for _ in range(cols)] for _ in range(rows)]
    return matrix
```